

TELECOM CUSTOMER RETENTION ANALYTICS

Machine Learning-Based Customer Churn Prediction & Retention Strategy

Author: Alin Kumar

Executive Summary

End-to-end telecom churn prediction project using EDA, feature engineering, model comparison, hyperparameter tuning and business recommendations.

Key Findings

Month-to-month customers showed highest churn. Two-year contracts showed strongest retention. Fiber optic and electronic check customers were higher risk.

Final Model

Balanced Logistic Regression selected with approximately 78% recall and strong business usefulness.

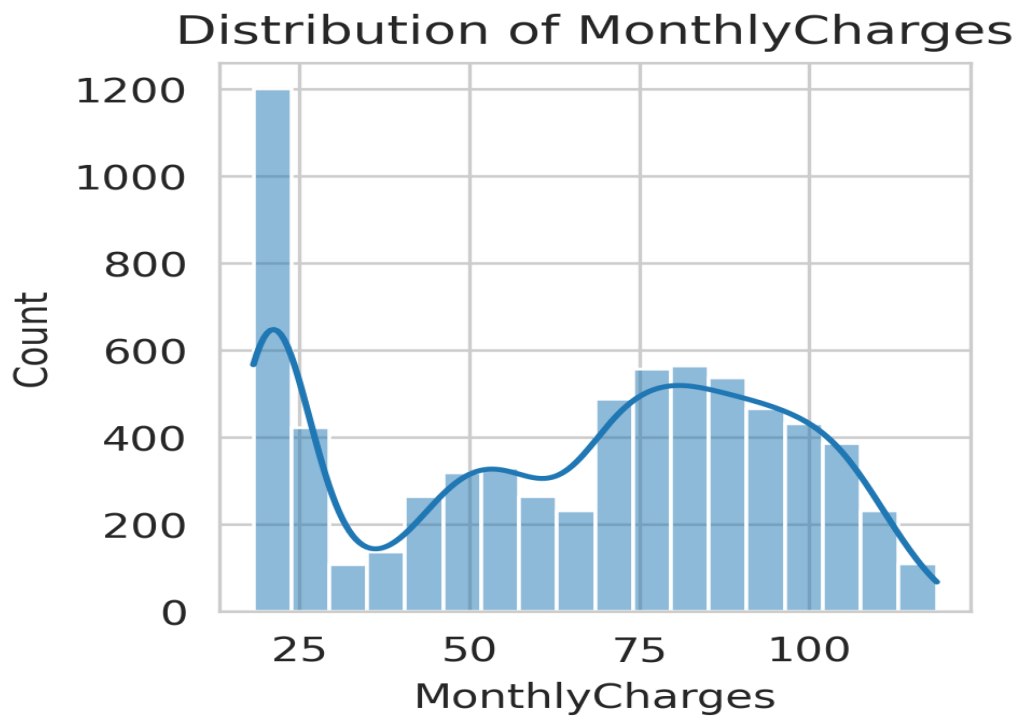
Business Recommendations

Promote annual plans, target low-tenure customers, improve fiber service experience, encourage auto-pay, and bundle security/support services.

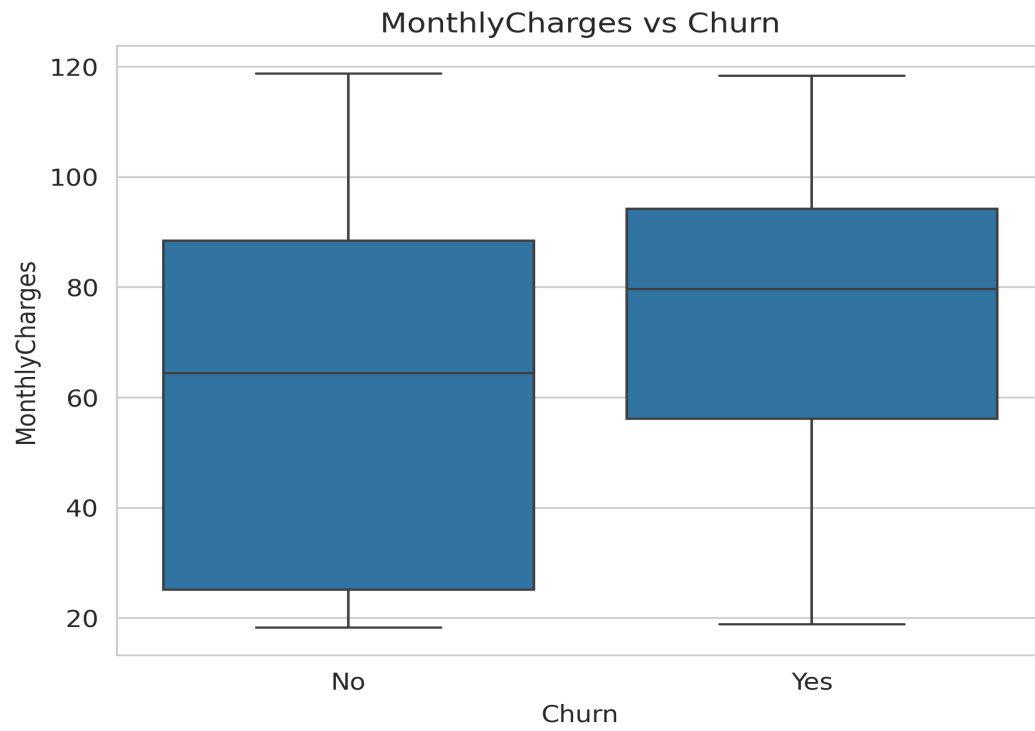
Feature Importance - Balanced Logistic Regression.png

InternetService_vs_churn.png

MonthlyCharges_distribution.png



MonthlyCharges_vs_churn.png



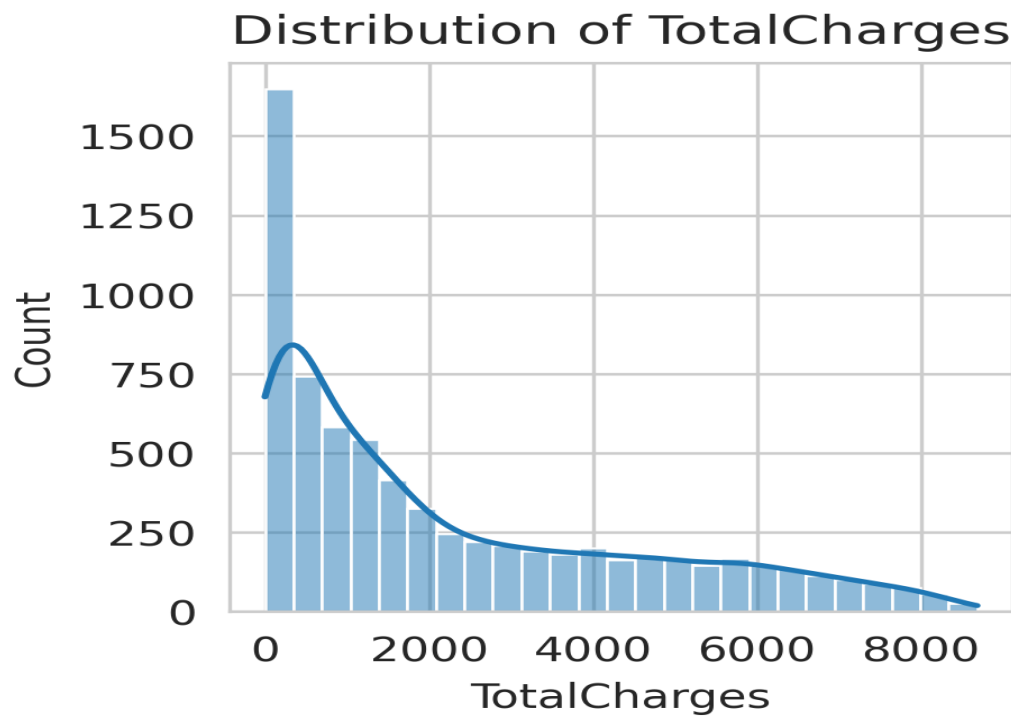
PaymentMethod_vs_churn.png

TechSupport_vs_churn.png

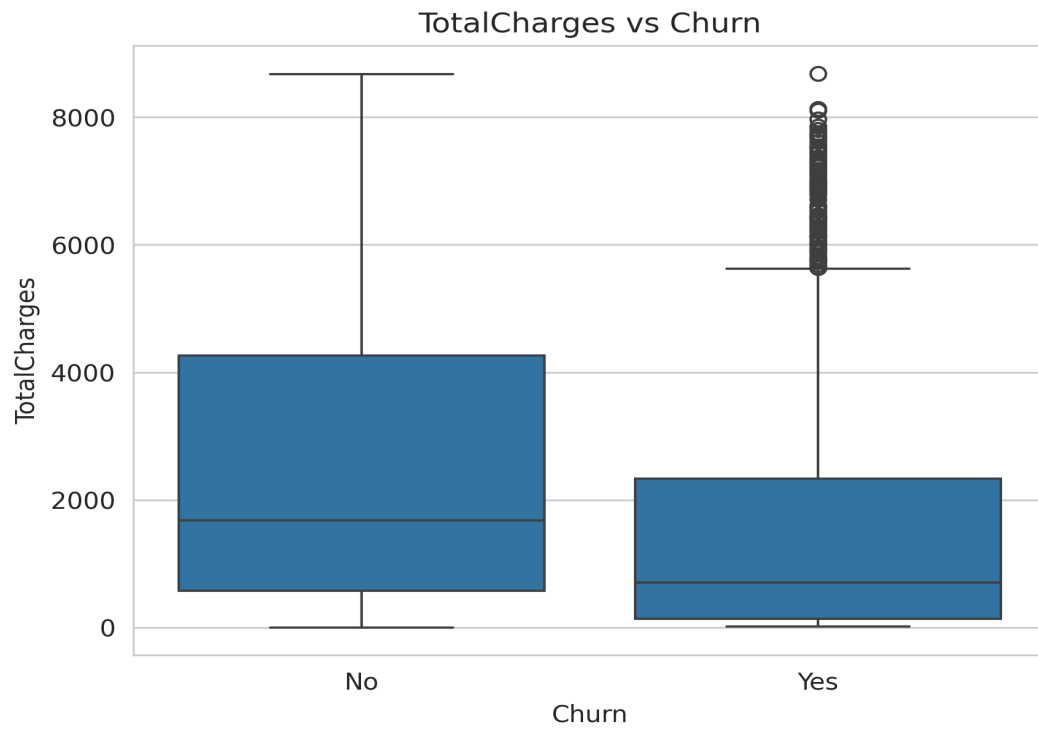
Top 10 Feature Increasing Churn.png

Top 10 Feature Reducing Churn.png

TotalCharges_distribution.png



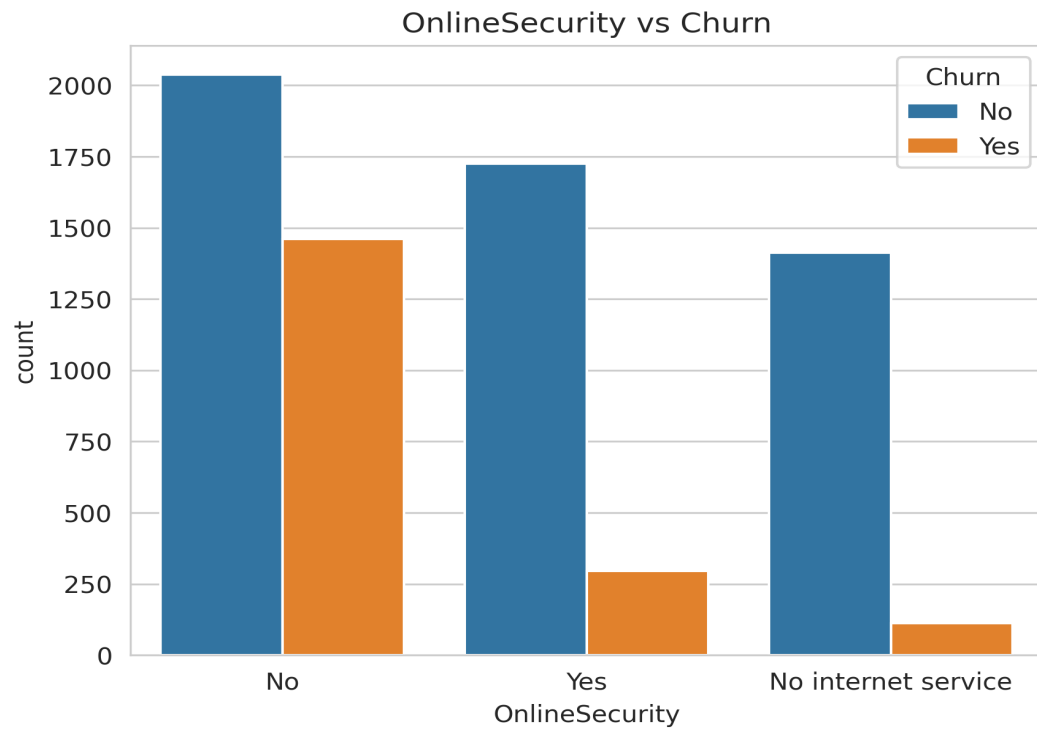
TotalCharges_vs_churn.png



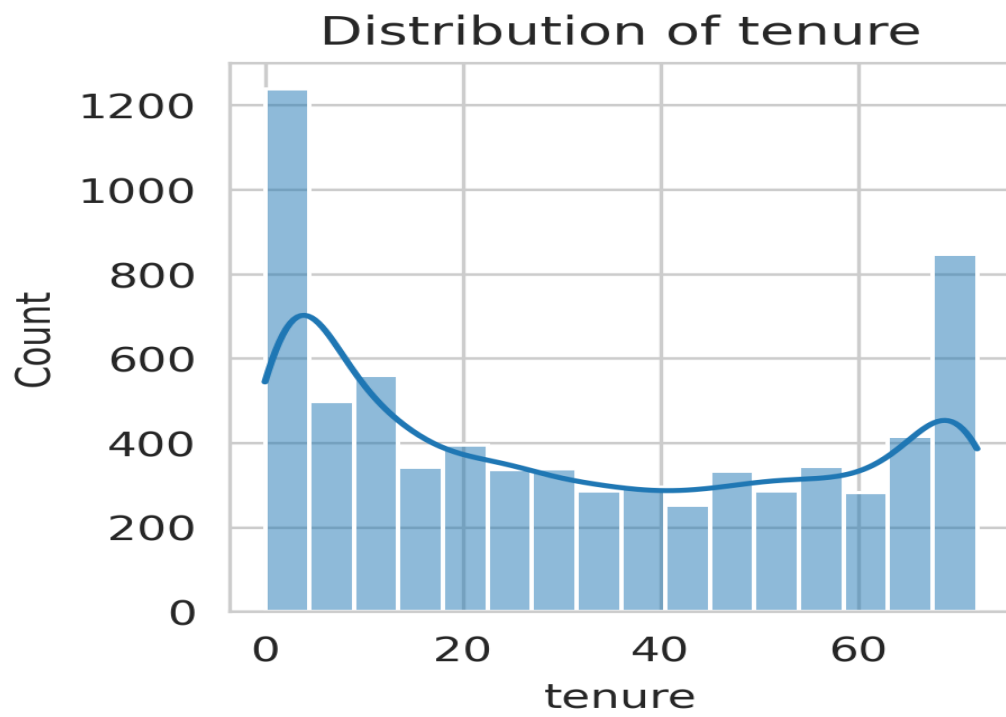
contract_vs_churn.png

correlationMatrix.png

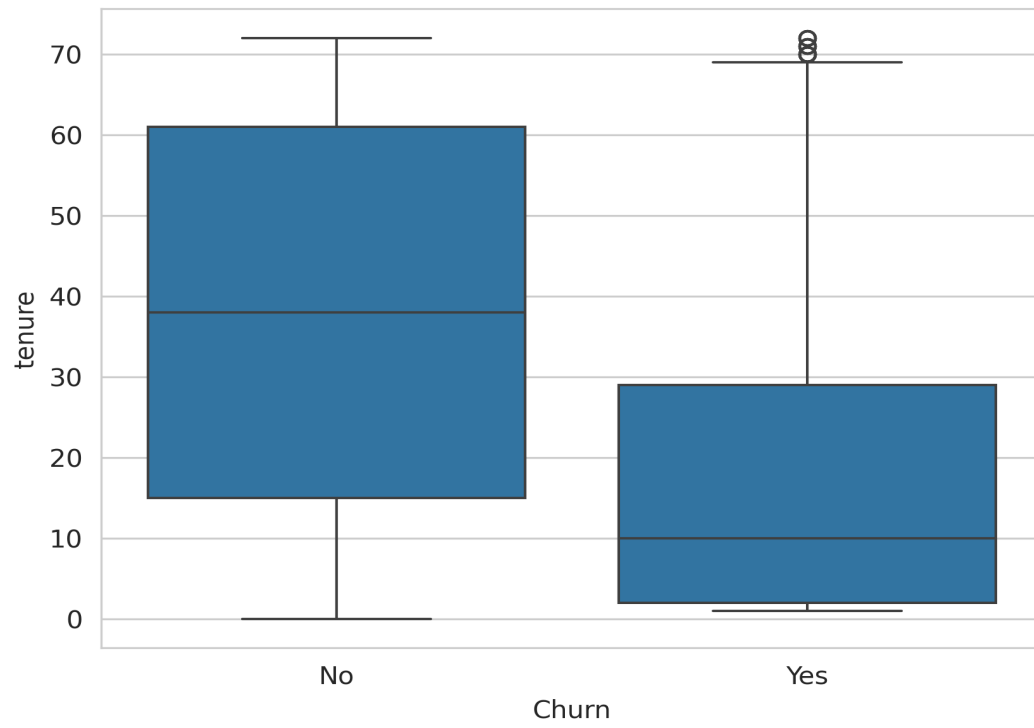
onlinesecurity_vs_churn.png



tenure_distribution.png



tenure_vs_churn.png



Conclusion

Data does not reduce churn. Action does. The true value of analytics lies in enabling organizations to act before valuable customers are lost.